

Candidate Data Transformer - Technical Design

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Pipeline

detect -> extract -> normalize -> deduplicate -> merge -> confidence -> project-to-output -> validate. Adapters convert ATS JSON, recruiter CSV, GitHub API snapshots, and recruiter notes into a common SourceCandidate shape. The engine then produces an internal canonical Candidate before any customer-specific projection is applied.

Canonical Schema And Normalization

The canonical record stores candidate_id, full_name, emails, phones, location, links, headline, years_experience, skills, experience, education, provenance, and overall_confidence. Emails are lowercased and validated, phones are normalized to E.164-style strings, countries become ISO alpha-2 codes, dates become YYYY-MM, and skill aliases such as py, postgres, and SpringBoot become canonical names.

Merge And Conflict Policy

Records are grouped by email, then phone, then name only when stronger keys are absent. Source priority is ATS JSON 0.90, recruiter CSV 0.85, GitHub 0.70, and recruiter notes 0.60. Scalar fields choose the highest-confidence available value; list fields are unioned deterministically. Every chosen or extracted value keeps provenance with source, method, and confidence.

Projection Config

The runtime config reshapes output without code changes. It can select fields, rename paths with from, apply E164 or canonical normalizers, include or hide confidence/provenance, and choose null, omit, or error behavior for missing values. Validation runs after projection so the returned JSON matches the requested shape.

Edge Cases

Case	Handling
Malformed source file	Adapter returns an empty list; the run continues.
Bad email/phone	Normalizer returns null instead of inventing a value.
Same person across sources	Deduped by email/phone and merged with provenance.
Conflicting names or titles	Higher-confidence source wins.
No usable identity	Skipped to avoid polluting hiring data.

Deliberately Descoped

Live GitHub calls and resume PDF/DOCX parsing are left out for determinism and focus. The architecture can add those as new adapters without changing merge, projection, or validation logic.